

A PROJECT REPORT ON COMPLAINT MANAGEMENT SYSTEM

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Submitted by

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ABSTRACT

The Complaint Management System is a web-based application designed to streamline the process of lodging, tracking, and resolving complaints within an organization. Traditional complaint handling methods rely heavily on manual recording through handwritten logbooks or informal communication channels such as emails or verbal reporting. These methods are error-prone, lack proper traceability, and often lead to unresolved grievances and user dissatisfaction.

This project proposes a centralized, digital solution that enables users to register complaints under predefined categories and track the progress of their submissions. Administrators can view active complaints, update their status, and provide resolution remarks using a structured dashboard. The system enforces secure authentication, role-based access control, and maintains a historical archive of all complaints for analysis and auditing.

The implementation is based on a three-tier architecture comprising a web-based user interface, an application server built using Node.js and Express, and a MySQL relational database. The system improves transparency, enhances accountability, and enables data-driven decision-making by providing insights into recurring issues and service bottlenecks.

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CHAPTER 1

INTRODUCTION

1.1 Background

In academic institutions, corporate offices, residential communities, and public service organizations, complaints and grievances are inevitable. These complaints may be related to technical issues, infrastructure problems, administrative delays, or service quality. Efficient handling of complaints is essential to maintain trust, improve operational efficiency, and ensure user satisfaction.

Many organizations rely on manual processes for complaint handling, such as maintaining physical registers, collecting feedback forms, or using unstructured communication through email or messaging platforms. These approaches lack a central repository, an automated tracking mechanism, and clear accountability, which often results in complaints being overlooked or poorly managed.

1.2 Need for the System

The absence of a structured complaint management solution leads to several challenges:

- Difficulty in tracking the lifecycle of a complaint from submission to closure.
- Lack of visibility for users regarding the current status of their complaints.
- Limited ability for administrators to analyze recurring complaints and trends.
- High dependency on manual effort, leading to delays and human errors.

A web-based Complaint Management System addresses these challenges by providing a centralized, transparent, and automated mechanism to handle complaints efficiently.

1.3 Project Definition

The Complaint Management System is a role-based web application that allows users to register and monitor complaints, and enables administrators to manage and resolve them. The system provides:

- A user interface for authentication, complaint submission, and history tracking.
- An admin interface for viewing, filtering, updating, and archiving complaints.
- A backend server that handles data processing, validation, and communication with the database.
- A database layer for persistent storage of user and complaint information.

1.4 Structure of the Report

This report is organized into multiple chapters. The introduction outlines the background, need, and definition of the project. The following chapters cover the problem statement, objectives, scope, system analysis, system design, module descriptions, implementation details, testing and validation, results, conclusion, and future enhancements.

CHAPTER 2

PROBLEM STATEMENT AND OBJECTIVES

2.1 Problem Statement

The existing manual or semi-manual complaint management methods suffer from several drawbacks:

- **Decentralized Storage:** Complaints are spread across disparate mediums (log-books, emails, verbal reports), increasing the likelihood of data loss.
- **Lack of Transparency:** Users do not have real-time visibility into the status of their complaints or the actions taken.
- **Limited Accountability:** It is difficult to determine responsibility for delayed or unresolved complaints. **Poor Traceability:** There is no structured mechanism to maintain historical data for analysis or auditing.
- **Redundancy and Inconsistency:** Multiple entries of the same complaint and inconsistent formats are common.

These issues highlight the need for a centralized and automated system that can effectively manage the complaint lifecycle.

2.2 Objectives

The primary objectives of the Complaint Management System are as follows:

- To provide an intuitive web-based interface for users to lodge complaints.
- To ensure secure authentication and role-based access control for users and administrators.
- To enable real-time status tracking and updates for each complaint.
- To provide administrators with a centralized dashboard to manage active and resolved complaints.
- To maintain a historical archive of complaints for analysis and performance evaluation.

- To minimize manual work, reduce delays, and improve overall efficiency.

2.3 Scope of the Project

The scope of this project includes the design and implementation of a functional prototype with the following features:

- User registration and login using unique email credentials.
- Role differentiation between normal users and administrators.
- Complaint submission with category selection and detailed description.
- User dashboard for viewing complaint history and current status.
- Admin dashboard for viewing active complaints, updating status, and adding resolution messages.
- Storage of user and complaint data in a relational database.

The project does not currently include features such as email notifications, file attachments, or mobile application support, which are considered as potential future enhancements.

CHAPTER 3

SYSTEM ANALYSIS

3.1 Existing System

In a conventional setup, complaints are often collected via paper forms, logbooks, emails, or informal channels. The existing system has the following disadvantages:

- High risk of misplacing or losing complaint records.
- No centralized database to store and organize data.
- Lack of automatic updates or feedback to users.
- Difficulty in prioritizing complaints and tracking unresolved cases.

3.2 Proposed System

The proposed Complaint Management System introduces a fully digital workflow:

- Users can securely log in and register complaints.
- Complaints are stored in a structured MySQL database.
- Admins manage complaints via a unified web interface.
- Status changes and resolution messages are visible to users.

This system enhances traceability, transparency, and accountability while significantly reducing the manual workload.

3.3 Functional Requirements

The core functional requirements include:

- User registration and login functionality.
- Complaint submission with mandatory fields.
- Admin view of all submitted complaints.
- Update of complaint status (Submitted, In Review, Resolved).
- Display of complaint history for each user.

3.4 Non-functional Requirements

Non-functional requirements describe quality attributes:

- **Usability:** The interface should be easy to navigate.
- **Performance:** The system should handle concurrent requests efficiently.
- **Security:** User authentication and data access must be protected.
- **Reliability:** Data should be stored reliably without loss.
- **Scalability:** The solution should be extendable to larger user bases.

3.5 Software and Hardware Requirements

Software Requirements

- Operating System: Windows / Linux
- Backend: Node.js (Express framework)
- Frontend: HTML5, CSS3, JavaScript
- Database: MySQL
- Tools: Visual Studio Code, Git, Browser (Chrome/Firefox)

Hardware Requirements

- Processor: Intel i3 or higher
- RAM: 2 GB or higher
- Storage: Minimum 1 GB free space
- Network: Internet connectivity for client-server interaction

CHAPTER 4

SYSTEM DESIGN

System design describes the overall architecture, data models, and interaction between components.

4.1 Architectural Design

The Complaint Management System follows a layered, web-based architecture comprising:

- Presentation Layer (Web UI)
- Application Layer (Node.js + Express)
- Data Layer (MySQL Database)

System Architecture Diagram

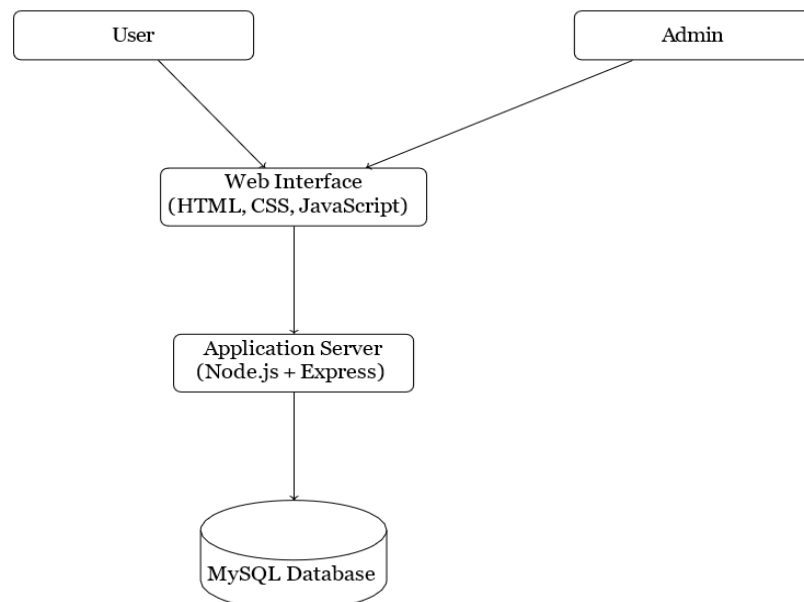


Figure 4.1: System Architecture of Complaint Management System

4.2 Data Flow Overview

The data flow can be summarized as follows:

- Users send HTTP requests from the browser to the server.
- The server validates input and interacts with the MySQL database.
- Responses such as status updates and complaint lists are sent back to the client.

4.3 User Interface Design

The user interfaces are designed using modern layout techniques and a glassmorphism-inspired visual style.

UI Screens

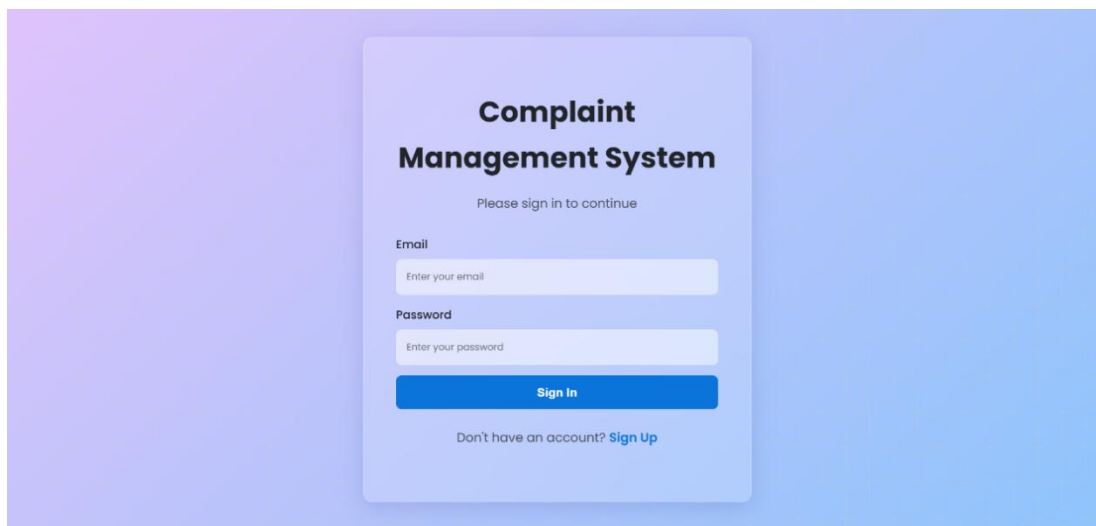


Figure 4.2: Sign In Interface

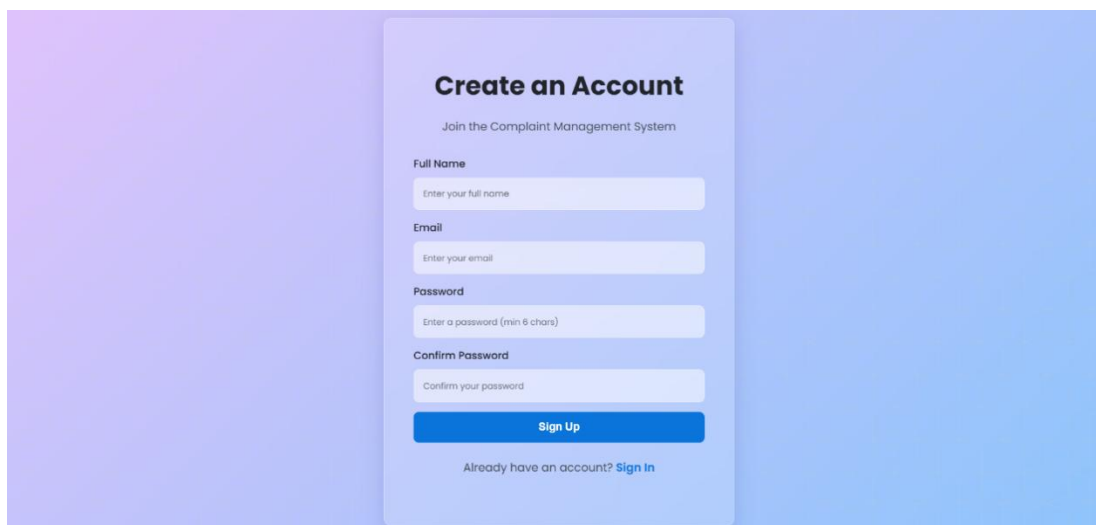


Figure 4.3: Sign Up Interface

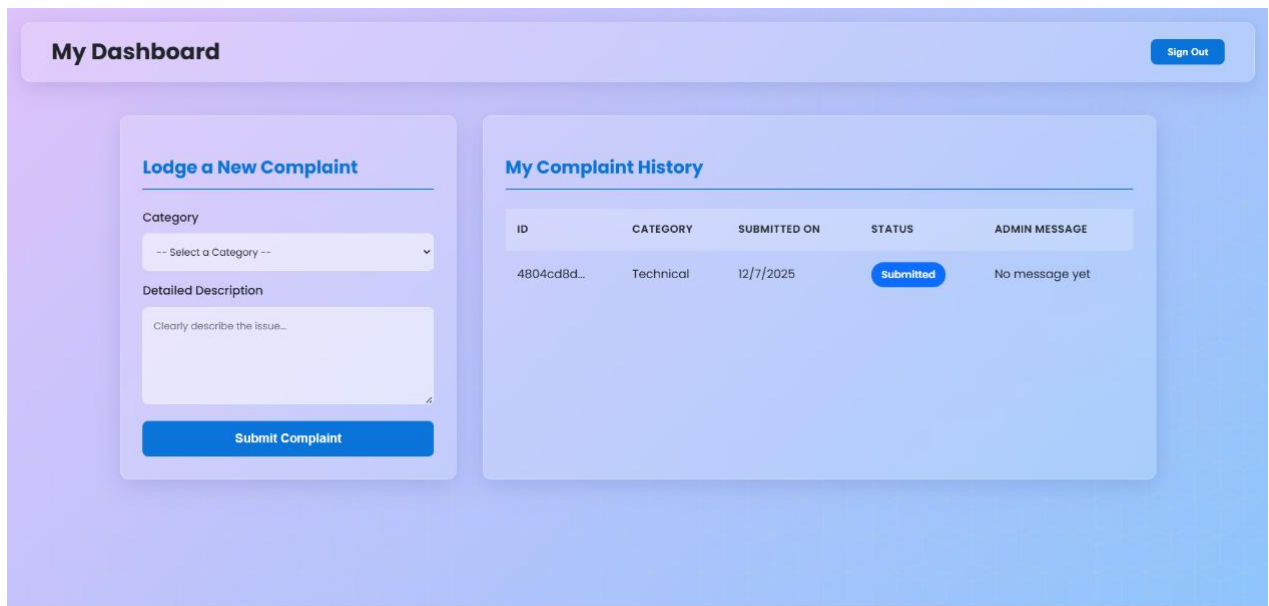


Figure 4.4: User Dashboard Interface

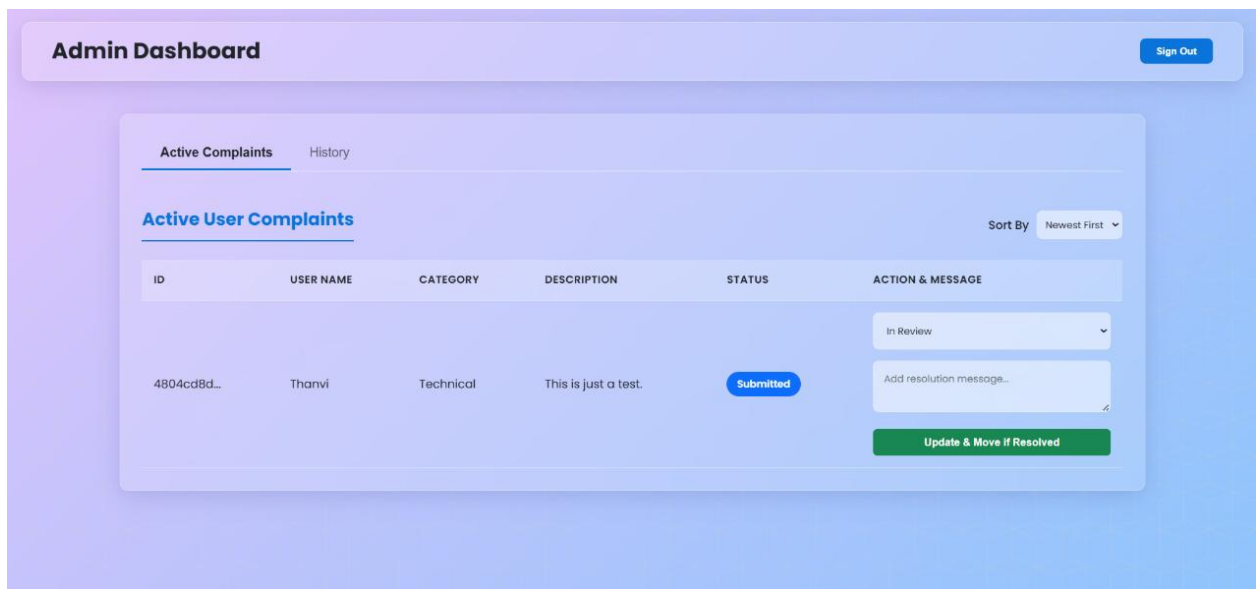


Figure 4.5: Admin Dashboard Interface

CHAPTER 5

MODULE DESCRIPTION

This chapter describes the major modules and their functionality.

5.1 Authentication Module

This module handles user registration and login. It includes:

- Registration with full name, email, and password.
- Password validation and hashing before storage.
- Login checks using stored credentials.
- Session management for authenticated access.

5.2 User Module

The user module provides features to:

- Submit new complaints by selecting a category and entering a description.
- View a personalized complaint history.
- Track complaint status and read admin resolution messages.

5.3 Admin Module

The admin module is responsible for:

- Viewing all active complaints in a tabular format.
- Updating complaint status from “Submitted” to “In Review” or “Resolved”.
- Providing resolution comments for each complaint.
- Viewing resolved complaints in a separate history tab.

5.4 Complaint Management Module

This module ensures:

- Each complaint is assigned a unique identifier.
- Status transitions are handled logically.
- Resolution messages are stored securely.

5.5 Security Module

Security-related aspects include:

- Password hashing using bcrypt.
- Session-based authentication.
- Separation of user and admin permissions.

CHAPTER 6

DATABASE DESIGN

The database is implemented using MySQL and follows a relational schema.

6.1 Schema Overview

The primary entities are:

- **User:** Represents a system user with login credentials and role.
- **Complaint:** Represents an individual complaint submitted by a user.

6.2 User Table

Field	Type	Constraint	Description
id	INT	Primary Key, Auto Auto Increment	Unique user identifier
fullName	VARCHAR(255)	Not Null	Full name of the user
email	VARCHAR(255)	Unique, Not Null	Login email address
password	VARCHAR(255)	Not Null	Hashed password
role	ENUM('user', 'admin')	Default 'user'	Role of the account

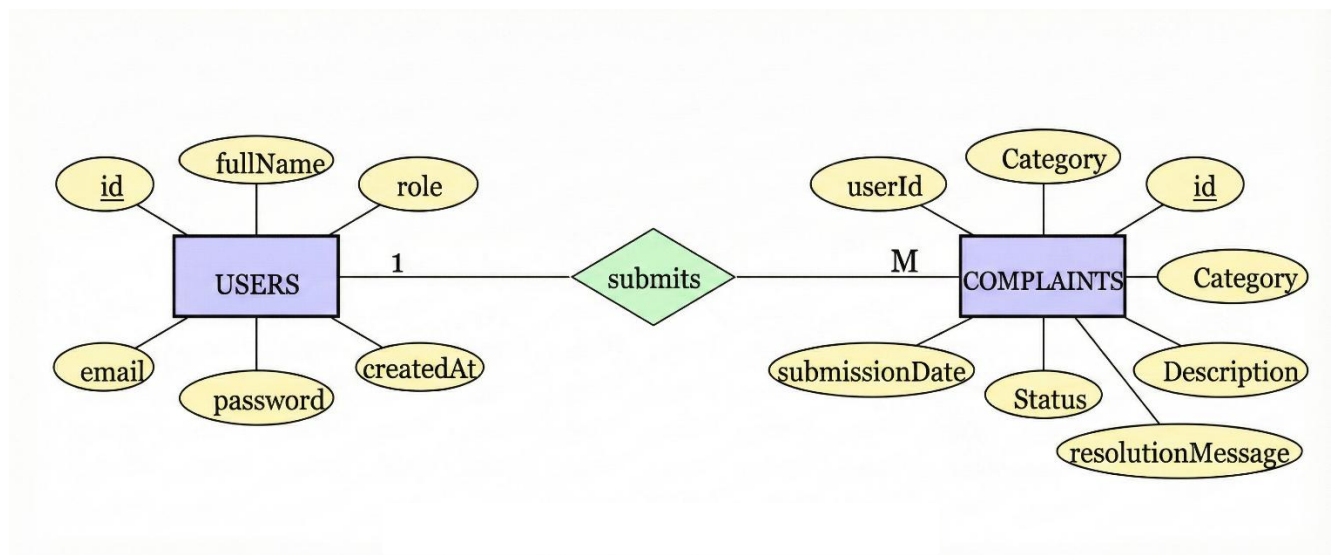
Table 6.2: User Table Structure

6.3 Complaint Table

Field	Type	Constraint	Description
id	CHAR(36)	Primary Key	Unique complaint ID (UUID)
userId	INT	Foreign Key	References user who created the complaint
Category	VARCHAR(100)	Not Null	Complaint category
Description	TEXT	Not Null	Detailed complaint description
Status	VARCHAR(50)	Default 'Submitted'	Current status of the complaint
submissionDate	TIMESTAMP	Default CURRENT_TIMESTAMP	Date and time of submission
resolutionMessage	TEXT	Nullable	Admin's resolution message

Table 6.3: Complaint Table Structure

6.4 ER Diagram



CHAPTER 7

IMPLEMENTATION DETAILS

7.1 Technology Stack

- Frontend: HTML5, CSS3, Vanilla JavaScript
- Backend: Node.js with Express framework
- Database: MySQL
- Libraries: bcrypt, express-session, mysql2, uuid

7.2 Backend Architecture

The backend is implemented using Express routes that handle:

- User sign-up and login (authentication).
- Complaint creation and retrieval.
- Admin-only endpoints for updating complaint status.

7.3 Security Considerations

- Passwords are always stored in hashed form using bcrypt.
- Sessions are managed using express-session, ensuring authenticated access.
- Admin operations are protected using role checks.

CHAPTER 8

TESTING AND RESULTS

8.1 Testing Strategy

The system was tested using:

- Unit testing of individual routes.
- Integration testing of full complaint submission and resolution flow.
- Manual UI testing for validation and error handling.

8.2 Sample Test Cases

ID	Test Scenario	Input	Expected Result	Status
TC1	User Registration	Valid name, email, password	Account created successfully	Pass
TC2	Invalid Login	Wrong password	Error message displayed	Pass
TC3	Complaint Submission	Valid category and description	Complaint Stored and visible in history	Pass
TC4	Admin Status Update	Change status to Resolved	Complaint moved to history tab	Pass

8.3 Results

The Complaint Management System met the functional and non-functional requirements defined during analysis. Users can register and log in, submit complaints, and view updates. Administrators can effectively track and resolve complaints using a centralized interface.

CHAPTER 9

ADVANTAGES AND LIMITATIONS

9.1 Advantages

- Centralized repository of complaints and user records.
- Real-time status visibility for users.
- Enhanced accountability and transparency.
- Reduced manual workload and paperwork.

9.2 Limitations

- No automated email or SMS notification system.
- No support for file attachment with complaints.
- Only a single admin role is considered in the current design.

CHAPTER 10

CONCLUSION AND FUTURE ENHANCEMENTS

10.1 Conclusion

The Complaint Management System successfully addresses the limitations of manual complaint handling methods by providing a structured, web-based solution. The system enhances transparency, traceability, and user satisfaction by enabling users to submit and track complaints while providing administrators with efficient management tools.

10.2 Future Enhancements

Potential future improvements include:

- Integration of email and SMS notifications.
- Support for uploading attachments such as images or documents.
- Advanced analytics and dashboard for complaint trends.
- Role-based multi-level administration.
- Mobile application version for Android and iOS.

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